

# Presentation-Level Universal Property of H-epsilon

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## Summary

Pass 73 proves the universal property currently available for  $\mathcal{H}_\epsilon$ .

The result is deliberately limited: it is a presentation-level universal property among support-preserving certificate targets. It is not yet a faithful embedding into LCA sheaves, condensed/solid abelian groups, or a canonical exact pro-category.

## Certificate Targets

A support-preserving certificate target for  $\epsilon_{\mathbb{P}}$  supplies images of five generator families:

1. finite conductor windows  $W_{S,k}$  and lattices  $L_{S,k}$ ;
2. Loeb-Rosser boundaries  $d_S$ ;
3. restriction maps for finite prime inclusions;
4. signed duality maps  $d_S \mapsto -d_S^T$ ;
5. the derived pro-Ab lcm tower  $K_n = N_n \mathbb{Z}$ .

It is admissible when these images satisfy the finite/pro relations already checked for  $\mathcal{H}_\epsilon$ : finite exactness, restriction composition, signed-dual compatibility, conductor bookkeeping, and non-Mittag-Leffler pro growth.

## Universal Property

Since  $\mathcal{H}_\epsilon$  is presented by exactly these generators and relations, every admissible support-preserving certificate target  $C$  receives a unique generator-preserving functor

$$\mathcal{H}_\epsilon \rightarrow C.$$

Thus  $\mathcal{H}_\epsilon$  is initial among admissible support-preserving certificate targets.

## Minimality

Each generator family is necessary:

- without conductor windows, local support-preserving duality is untyped;
- without boundaries  $d_S$ , the finite class  $\epsilon_S$  is undefined;
- without restrictions, finite shadows do not assemble to  $\epsilon_{\mathbb{P}}$ ;
- without signed duality, the functional equation is untyped;
- without the lcm tower, finite CRT levels remain zero and  $\widehat{\mathbb{Z}}/\mathbb{Z}$  is lost.

## Verification

The checker `code/scripts/check-pass73.py` produces `artifacts/reports/pass73-h-epsilon-universal-property-check.json`.

It verifies:

- finite conductor normal forms through six primes and  $k \leq 3$ ;
- pro normal forms through  $N_{24}$ ;
- restriction and signed-dual relations;
- lcm-tower cofinality and non-Mittag-Leffler growth;
- unique generator-preserving functors into admissible targets;
- obstruction witnesses for targets omitting any generator family.

The remaining problem is external realization: construct a faithful exact functor from  $\mathcal{H}_\epsilon$  into LCA sheaves, condensed/solid groups, or an exact pro-category with restricted-product generators, or prove a no-go theorem for such a realization.